

Qatar-1b

R-0251

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180506 · created 2026-07-17T06:38:51+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458244.8575 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.011
Transit depth (Rp/Rs)^2	1.33 +/- 0.26 [%]
Orbital Inclination (inc)	84.52 +/- 0.71
Airmass coefficient 1 (a1)	1.325 +/- 0.016
Airmass coefficient 2 (a2)	-0.1931 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	1.21 %
Best Comparison Star	None
Optimal Aperture	3.48
Optimal Annulus	7.2
Transit Duration (day)	0.0685 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.011 vs 0.1463 (deviation 2.26 σ)
Duration fit vs published	0.0685 d vs 0.0692 d (deviation 0.15 σ)
Mid-time O-C	-0.3 min
Depth SNR	8.3
Residual scatter	1.21 %

Light curve

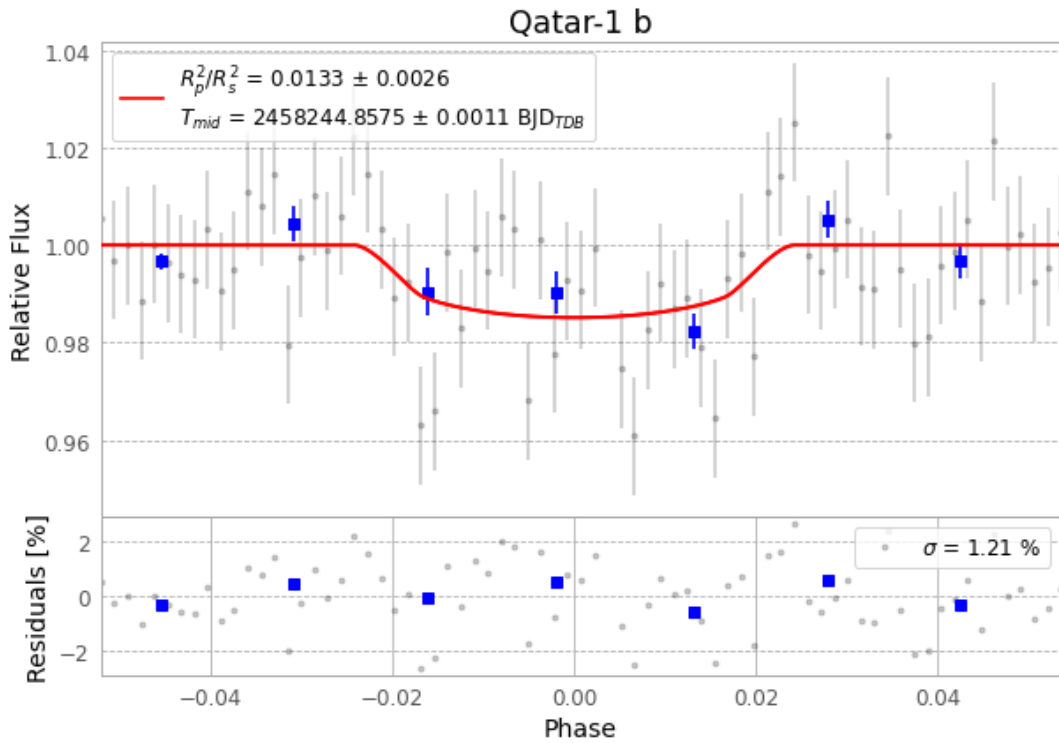


Figure 1 — FinalLightCurve_Qatar-1 b_2018-05-05.png (SHA-256 1ffd2d42dcee30d4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [436, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aab86c00a41b2975...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), Qatar-1180506064812.FITS → Qatar-1180506102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180506.log (0aa32bfd7e36...), FinalLightCurve_Qatar-1 b_2018-05-05.png (1ffd2d42dcee...), FinalLightCurve_Qatar-1 b_2018-05-05.csv (339502e60345...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 06:38Z.